

Software Requirements Specification

for AI-Assisted Smart Campus Placement & Career Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Project Team	03-01-2026	Initial Draft	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the **AI-Assisted Smart Campus Placement & Career Management System**. This document is intended to be used by developers, testers, and stakeholders to understand the system's purpose, features, constraints, and interfaces.

1.2 Scope

The purpose of this system is to automate and centralize campus placement activities including student profiling, placement drive management, eligibility checking, and career analytics. The system will assist students and administrators in managing placement-related workflows efficiently.

1.3 Overview

This SRS includes sections detailing product perspective, functionality, user characteristics, external interfaces, and the detailed functional and non-functional requirements of the system in a structured IEEE format.

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
 - *AI-Assisted Smart Campus Placement & Career Management System* Vision Document (if available).
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2. Overall Description

2.1 Product Perspective

The *AI-Assisted Smart Campus Placement & Career Management System* is a standalone web-based application aimed at replacing manual placement management practices like spreadsheets and email workflows. It includes intelligent features such as eligibility scoring and analytics to support decision-making.

2.2 Product Functions

High-level functions include:

1. **Student Profile Management:** profile creation and updates
2. **Placement Drive Creation and Scheduling:** tools for admins to manage drives.
3. **Eligibility Calculation & Recommendation:** automated checks against company criteria.
4. **Application Process and Status Tracking:** real-time tracking of applications.
5. **Career Insights Analytics:** data-driven insights for students and admins.

(Detailed functional requirements are in Section 4)

2.3 User Classes and Characteristics

User Class	Description	Technical Skill
Student	End user applying for drives	Basic
Admin/TPO	Placement coordinator	Intermediate
Recruiter	Company HR accessing drive details	Intermediate

2.4 Operating Environment

The system will operate in a web environment using standard web browsers (Chrome, Firefox, Edge) on desktop or mobile devices. Backend server requirements and DBMS will be specified later during design.

2.5 Design and Implementation Constraints

- The system backend shall be developed using Java (Spring Boot or equivalent framework) to ensure scalability and maintainability.
- Python shall be used exclusively for implementing AI-assisted modules such as eligibility scoring and skill analysis.

- The frontend shall be implemented using React.js or Next.js to provide a responsive and interactive user interface.
- A relational database (MySQL / PostgreSQL / MongoDB) shall be used for persistent data storage.
- Communication between frontend, backend, and AI modules shall be handled through RESTful APIs.
- The system must adhere to secure coding practices including authentication, authorization, and data encryption.
- Future versions may extend the AI module to machine learning-based recommendations without impacting the core system architecture.

2.6 User Documentation

A user guide/manual and online help will be provided covering login, profile management, drive application, and analytics dashboards.

2.7 Assumptions and Dependencies

- Internet connection availability for all users.
 - Students have valid login credentials provided by the institution.
 - Recruitment data supplied by companies is accurate.
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3. External Interface Requirements

3.1 User Interfaces

The system will have the following UI screens:

Login Page: Secure access point.

Student Dashboard: Personalized view for students.

Admin Dashboard: Overview and controls for TPOs.

Placement Drive Listing: Searchable list of opportunities.

Application Status View: Tracking for applied jobs.

3.2 Hardware Interfaces

The system will not interact with external hardware devices. It will run on standard web servers.

3.3 Software Interfaces

Interfaces to: * **Database Server:** MySQL/SQL Server for persistent storage. * **Web Server:** To host the application. * **Email System:** For sending notifications (SMTP).

3.4 Communications Interfaces

- **HTTP/HTTPS:** Protocols for client-server communication.
 - **REST APIs:** Used internally between front-end and back-end components.
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4. System Features

4.1 Student Profile Management

4.1.1 Description and Priority

This feature allows students to register, login, and update their academic and career details.

Priority: High

4.1.2 Stimulus/Response Sequences

- Student enters profile data -> System saves & validates data.

4.1.3 Functional Requirements

- **REQ-SPM-001:** The system shall allow students to register an account.
- **REQ-SPM-002:** The system shall let students update their profile information including GPA, skills, and resume.

4.2 Placement Drive Management

4.2.1 Description and Priority

Allows Admin to create and manage placement drives.

Priority: High

4.2.2 Stimulus/Response Sequences

- Admin posts a new drive -> System notifies eligible students.

4.2.3 Functional Requirements

- **REQ-PDM-001:** The system shall enable admin to create placement drives with criteria (cutoff, branch, etc.).
- **REQ-PDM-002:** The system shall allow drive details to be updated.

4.3 Eligibility Scoring Engine

4.3.1 Description and Priority

Automated eligibility scoring using rule-based logic.

Priority: High

4.3.2 Stimulus/Response Sequences

- Student views a drive -> System calculates and displays eligibility status.

4.3.3 Functional Requirements

- **REQ-ESE-001:** The system shall compute eligibility score for students based on academic records.
- **REQ-ESE-002:** The system shall recommend eligibility status (Eligible/Not Eligible).

Note: The eligibility scoring is implemented using rule-based logic and weighted criteria. This system is AI-assisted and does not involve complex machine learning models in the current version.

4.4 Application & Status Tracking

4.4.1 Description and Priority

Allows students to apply and track their progress.

Priority: High

4.4.2 Stimulus/Response Sequences

- Student clicks Apply -> System records application.

4.4.3 Functional Requirements

- **REQ-AST-001:** The system shall record student applications for specific drives.
- **REQ-AST-002:** The system shall display application status (Applied, Shortlisted, Rejected, Offered) to users.

4.5 Career Insights & Analytics

4.5.1 Description and Priority

Provides visual insights into placement trends.

Priority: Medium

4.5.2 Stimulus/Response Sequences

- Admin requests report -> System generates visual graphs.

4.5.3 Functional Requirements

- **REQ-CIA-001:** The system shall generate career analytics reports (placement stats, branch-wise selection, etc.).
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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support at least **100 concurrent users** with an average response time of ≤ 2 **seconds** under normal operating conditions.

5.2 Safety Requirements

- Data backups must be performed regularly to prevent data loss.

5.3 Security Requirements

- User authentication required for all access.
- Role-based access control (RBAC) to separate Admin and Student capabilities.

5.4 Software Quality Attributes

- **Usability:** Intuitive interface for non-technical users.
- **Reliability:** High availability (99.9% uptime).
- **Maintainability:** Modular code structure.

5.5 Business Rules

- Only registered students with verified emails can apply for placement drives.
 - Companies must be approved by the TPO before their drives are visible to students.
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6. Other Requirements

6.1 Database Requirements

- **Entities:** Student table, Drive table, Application table, Eligibility score data, Admin table.

6.2 Legal/Compliance Requirements

- Data privacy policies to be followed (GDPR/Local Data Protection Acts where applicable).
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Appendix A: Glossary

- **SRS** – Software Requirements Specification
- **TPO** – Training & Placement Officer
- **RBAC** – Role-Based Access Control

Appendix B: Analysis Models

Appendix B will include Use Case Diagrams, Class Diagrams, Sequence Diagrams, and State Diagrams prepared in subsequent practicals.

Appendix C: To Be Determined List

- Specific server hardware configuration (TBD).
- Integration with external legacy university database (TBD).